# FIBRES OF A SURJECTIVE LOCALLY FINITE TYPE MORPHISM FROM AN IRREDUCIBLE SCHEME TO THE SPECTRUM OF A VALUATION RING ARE EQUIDIMENSIONAL

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The reduction step and the reason it is harmless are both written out by the author; nothing is referred to the reader.

*Editorial note.* The counted spans carry 1 navigation stub and no bibliography citation.

*Editorial note.* The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

*Editorial note.* The label lemma-dominate-valuation-ring-dimension-fibres is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

*Allowed standard prerequisites (editorial).*

- Dimension behaviour of a finite type domain over a valuation ring. Algebra, Lemma algebra-lemma-finite-type-domain-over-valuation-ring-dim-fibres, invoked at proof lines 3663-3664. Supplies the statement in the integral case.
- Passing to the reduction preserves the dimensions involved. Standard, used at proof lines 3660-3661. Supplies the reduction to the integral case.

*Source reference.* Original lemma 00QK: sources/stacks-chow/tags.txt, source lines 970–970. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

**Lemma 0.1.** *Let $f : X \to \mathrm{Spec}(R)$ be a morphism from an irreducible scheme to the spectrum of a valuation ring. If $f$ is locally of finite type and surjective, then the special fibre is equidimensional of dimension equal to the dimension of the generic fibre.*

*Proof.* We may replace $X$ by its reduction because this does not change the dimension of $X$ or of the special fibre. Then $X$ is integral and the lemma follows from Algebra, Lemma 00QK. □

*Date*: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.